Determining Language Grounding Failures in Vision-Language-Action Models

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Abstract

Vision-Language-Action (VLA) models report high aggregate success rates on manipulation benchmarks, but these single-number metrics hide whether models genuinely follow instructions or exploit visual shortcuts and memorized scene layouts. In this paper, we isolate the language channel using a scene-fixed causal evaluation protocol on `LIBERO-Goal`, holding the initial simulator state strictly constant while perturbing instructions across three architectures spanning a $16\times$ parameter range (SmolVLA-450M, OpenVLA-7B, and OpenVLA-OFT-7.5B).

We find that language is causally necessary and scale-invariant: destroying the instruction by blanking or scrambling it collapses task success across all scales. However, we show that widely reported word-class asymmetries—where models appear to bind object nouns strictly while treating action verbs as near-optional—are confounded by benchmark information structure. In `LIBERO-Goal`, the noun set uniquely identifies all ten tasks ($H(\text{task} \mid \text{nouns}) = 0$ bits), rendering the verb redundant. When we test genuine verb antonyms, OpenVLA-7B drops 31.7 points ($p < 10^{-5}$), proving that it reads the verb when the benchmark requires it. We present an episode-level goal predicate taxonomy to distinguish instruction-following from scene-memorized defaults, and argue that benchmark information content must be reported before declaring models instruction-blind.

1 Introduction

Vision-Language-Action (VLA) models combine vision-language backbones with robotic control heads, promising policies that execute physical manipulations driven by natural-language instructions (Brohan et al., 2022, 2023; Driess et al., 2023; Kim et al., 2024; Shukor et al., 2025). Standard evaluation benchmarks report high aggregate success rates, and the community frequently interprets these figures as evidence that policies genuinely understand their instructions. However, a model can score well on a benchmark without parsing language at all by exploiting visual shortcuts, memorized task layouts, or superficial keyword matching. This gap between aggregate success and genuine language grounding is the central problem we address in this paper.

We approach this problem through two core research questions:

- **RQ1:** Does language causally drive a VLA model's physical behavior when the initial visual scene is held strictly constant?
- **RQ2:** Which linguistic operations break grounding, and do observed failures stem from object reference (nouns) or action reference (verbs)?

To answer these questions, we hold the physical simulator scene fixed and treat only the instruction text as the experimental variable. This design choice separates our work from existing VLA robustness benchmarks, which vary multiple factors simultaneously—such as object positions, camera angles, lighting, sensor noise, and language—or compound several perturbations at once. By isolating language as the sole manipulated variable, we determine the instruction's exact role in driving policy behavior rather than making an aggregate robustness claim.

Evaluating three model architectures spanning a $16\times$ parameter range on `LIBERO-Goal`, we find that language is causally load-bearing and scale-invariant: destroying the instruction via blanking or length-matched nonsense collapses task success across all model scales. However, when we investigate word-class sensitivity, we discover that the widely reported asymmetry—where models seem sensitive to object nouns but insensitive to action verbs—is an artifact of benchmark information structure. In `LIBERO-Goal`, the noun set carries all 3.32 bits of task identity ($H(\text{task} \mid \text{nouns}) = 0$ bits), rendering the verb fully redundant. When we isolate the three tasks with true verb antonyms, OpenVLA-7B loses 31.7 points ($p < 10^{-5}$), demonstrating that it reads the verb when the benchmark requires it. We propose an episode-level goal-predicate taxonomy and metric suite to separate genuine instruction compliance from scene-memorized defaults.

2 Preliminaries

2.1 Vision-Language-Action Models

The VLA paradigm originated with RT-1 (Brohan et al., 2022), which framed robotic control as a sequence-modeling problem over tokenized actions. RT-2 (Brohan et al., 2023) showed that web-scale vision-language pre-training transfers into control, yielding emergent semantic understanding, while PaLM-E (Driess et al., 2023) demonstrated that a unified multimodal model benefits from joint training across language, vision, and control. Open X-Embodiment (Collaboration et al., 2023) scaled training data across more than a million real trajectories. These foundational works establish an expectation: if a policy inherits a language model's pre-trained semantics, it should condition its physical actions on the instruction. This expectation is precisely what our study puts to the test.

2.2 Evaluated Models

We evaluate three open-source checkpoints that instantiate the primary architectural choices in current VLA research:

1. **SmolVLA-450M** (Shukor et al., 2025): A compact policy combining a SmolVLM2 backbone with a *flow-matching* action expert, serving as our small-scale baseline.

2. **OpenVLA-7B** (Kim et al., 2024): The canonical 7B VLA policy built on Prismatic VLM and Llama-2, emitting *discretized* action tokens from a single static camera.

3. **OpenVLA-OFT-7.5B** (Kim et al., 2025): The Optimized Fine-Tuning recipe utilizing continuous action chunking, an L1-regression loss, an additional wrist camera stream, and 8-dimensional proprioception. (Note that OFT here refers to *Optimized* Fine-Tuning, not orthogonal fine-tuning; the checkpoint is evaluated without FiLM.)

These three models span a $16\times$ parameter range and three distinct action parameterizations (flow-matching, discretized tokens, and continuous L1 regression), allowing us to test whether language grounding behavior generalizes across architectures.

2.3 Benchmarks

We conduct all evaluations within the LIBERO benchmarking framework (Liu et al., 2023). LIBERO contains four distinct task suites: `Spatial`, `Object`, `Goal`, and `Long`. We focus exclusively on `LIBERO-Goal` because its ten tasks share an identical initial scene layout and object inventory. In `LIBERO-Goal`, the initial visual state S_0 does not disambiguate the goal, making the instruction the sole distinguishing input.

3 Related Work

VLA Robustness and Benchmark Shortcuts. Recent studies highlight severe vulnerabilities in VLA policies. Pang et al. (2025) evaluate the positional robustness of OpenVLA on `LIBERO-Object`, finding that success drops sharply outside the specific object positions seen during training, indicating rote memorization. LIBERO-Plus (Fei et al., 2025) benchmarks over 10,000 tasks across seven variation axes, reporting that language perturbations cause minor performance drops while visual perturbations disrupt policies severely. LIBERO-Para (Kim et al., 2026) evaluates paraphrase robustness across object and action references and introduces the PRIDE metric, which we adopt to measure paraphrase sensitivity.

The Instruction-Blindness Consensus. A growing literature argues that VLAs largely ignore language. LIBERO-PRO (Zhou et al., 2025) reports that outputs remain unchanged under corrupted instructions, attributing success to memorized action sequences. LangGap (Hou and Zhao, 2026) demonstrates that high-scoring models fail when forced to use language. Zhang and Bisk (2026) term this failure mode "instruction blindness", while Zhan et al. (2026) report a "modality collapse" toward visual affordances. Mechanistically, Grant et al. (2026) show that injecting baseline activations into null-prompt episodes recovers baseline behavior across six models.

Linguistic Shortcuts in NLP. Reliance on superficial dataset shortcuts is a well-documented challenge in computational linguistics. In Natural Language Inference (NLI), models routinely exploit lexical overlap and syntactic heuristics (Gururangan et al., 2018; McCoy et al., 2019). In Visual Question Answering (VQA), models frequently exploit language priors to answer without inspecting image details (Goyal et al., 2017). Our work connects this NLP dataset-artifact literature to VLA evaluation, demonstrating that apparent "verb blindness" in VLAs is an artifact of information redundancy in benchmark instruction design.

4 Language Grounding Failures

4.1 Defining Language Grounding Failure

We define a *language grounding failure* as an episode in which the policy's physical behavior is not causally driven by the linguistic content of the instruction. We distinguish this from an *execution failure*, in which the policy correctly infers the intended goal but fails to physically execute the trajectory.

Scalar success rates conflate these two independent properties: whether the policy understood the instruction, and whether the policy possessed the motor skill to complete it. A model can fail a task despite perfect language grounding (producing a correct approach trajectory that slips at the grasp phase). Conversely, a model can succeed at a task despite zero language grounding if its behavior is driven by memorized scene layouts or visual priors.

We resolve this ambiguity by holding the initial simulator state S_0 strictly constant while perturbing the instruction I. We then evaluate the final simulator state against all goal predicates achievable in that scene, rather than relying solely on the single predicate associated with the original task.

4.2 The Perturbation Taxonomy

Let I_{orig} denote the canonical instruction for a task on fixed scene S_0. We construct a family of perturbed instructions I_{cond} grouped by the shortcut hypothesis each condition falsifies:

- **Destructive Conditions:** Remove instruction content entirely while leaving scene S_0 untouched. `blank` supplies an empty string; `nonsense` supplies length-matched out-of-domain tokens. If the policy relies on language, success must collapse. If the policy succeeds, its behavior is driven by scene memorization.
- **Misleading Conditions:** Substitute a valid, executable instruction describing a different goal in S_0. `wrong_object` replaces the target object noun; `wrong_action` replaces the action verb; `wrong_task` replaces the full instruction with another task's instruction. These conditions test whether a policy follows the new instruction (compliance) or defaults to the scene's original goal (memorization).
- **Control & Paraphrase Conditions:** `repeated` concatenates I_{orig} with itself to test sensitivity to length. Paraphrase conditions (`para_object`, `para_action`, `para_compositional`) reword the instruction along specific linguistic axes while preserving meaning, testing robustness to surface form.

Table 1 summarizes the perturbation taxonomy.

4.3 Ground-Truth Evaluation via Goal Predicates

Standard simulators evaluate rollouts using a single boolean flag corresponding to the original task predicate g_{k^*}. Under perturbed instructions, this creates a mismatch: a policy that correctly follows a misleading instruction (e.g., executing `wrong_task`) is mis-scored as a failure by the native simulator check.

We exploit a structural property of `LIBERO-Goal`: all ten tasks are defined over the identical initial scene S_0. We evaluate every task's goal predicate g_k against the final state S_T of every rollout, constructing the *achieved task set*:

$$\mathcal{A}(S_T) = \{k : g_k(S_T) = \text{true}\} \quad (1)$$

where $k \in \{1, \ldots, 10\}$. $\mathcal{A}(S_T)$ reveals whether the policy achieved no goal ($\emptyset$), the original goal (k^*), or a newly instructed goal (k_{wrong}).

4.4 Evaluating Perturbed Instructions via Achieved Goal Sets

Using $\mathcal{A}(S_T)$, we establish per-episode classification schemes for destructive and misleading per-

Condition	Construction	Shortcut Hypothesis Ruled Out
`original`	I_{orig}, unchanged	Baseline reference.
`blank`	Empty string	Policy succeeds without language (pure scene memorization).
`nonsense`	Length-matched out-of-domain tokens	Policy relies on keyword presence without structural semantics.
`wrong_object`	Swapped target object noun	Policy ignores named object and defaults to a fixed target.
`wrong_action`	Swapped action verb	Policy ignores named action and defaults to a fixed manipulation.
`wrong_task`	Full instruction of another in-scene task	Policy ignores instruction and executes scene's default task.
`repeated`	I_{orig} concatenated with itself	Sensitivity to input length rather than meaning.
`para_object`, `para_action`, `para_compositional`	Reworded instruction preserving original goal	Policy is brittle to surface form despite unchanged semantics.

Table 1: The perturbation taxonomy: instruction construction and the specific shortcut hypothesis each condition rules out.

turbations.

4.4.1 Destructive Conditions

For `blank` and `nonsense`, no valid goal is specified. We cross whether I_{orig} succeeded against whether $\mathcal{A}(S_T) \neq \emptyset$ under the destructive condition (Table 2):

- **Grounded:** I_{orig} succeeded and $\mathcal{A}(S_T) = \emptyset$. Removing language eliminated success.
- **Memorization:** I_{orig} succeeded and $\mathcal{A}(S_T) \neq \emptyset$. The policy completed a task without language.
- **Spurious Success:** I_{orig} failed but $\mathcal{A}(S_T) \neq \emptyset$ under destructive prompt.
- **Uninformative:** I_{orig} failed and $\mathcal{A}(S_T) = \emptyset$. The policy could not complete the task under any prompt.

We define the episode-level Grounding Rate (GR) as:

$$\text{GR}(c) = \frac{|\text{Grounded}|}{|\text{Grounded}| + |\text{Memorization}| + |\text{Spurious}|} \quad (2)$$

4.4.2 Misleading Conditions

For `wrong_object`, `wrong_action`, and `wrong_task`, the perturbed prompt specifies goal $k_{\text{wrong}} \neq k^*$. We classify final states into three outcomes (Table 3):

- **Follow:** $k_{\text{wrong}} \in \mathcal{A}(S_T)$. The policy executed the newly instructed task (instruction compliance).
- **Ignore:** $k^* \in \mathcal{A}(S_T)$ and $k_{\text{wrong}} \notin \mathcal{A}(S_T)$. The policy executed the original task despite the altered prompt (scene-prior override).
- **Fail:** Neither goal achieved.

4.5 Metrics

We convert episode classifications into four aggregate metrics:

1. **Causal Sensitivity Score (CSS):** For destructive conditions, $\text{CSS}(c) = \frac{\text{TSR}_0 - \text{TSR}(c)}{\max(\text{TSR}_0, \epsilon)}$. CSS $\approx$ 1 indicates strong causal necessity; CSS ≈ 0 indicates language is ignored.
2. **Original-Action Rate (OAR):** For misleading conditions, $\text{OAR}(c) = \frac{|\text{Ignore}|}{|\text{Follow}|+|\text{Ignore}|+|\text{Fail}|}$. High OAR measures scene-memorization overrides.
3. **PRIDE Score:** For paraphrases, we compute task success weighted by linguistic paraphrase distance $\text{PD} = 1 - 0.5(\text{SK} + \text{ST})$, where SK and ST measure keyword and structural similarity (Kim et al., 2026).
4. **Failure Locus:** Failed episodes are categorized into *planning failures* (end-effector never approaches target within distance threshold τ) versus *execution failures* (end-effector reaches target but fails manipulation).

5 Experiments

5.1 Models and Benchmark

We evaluate SmolVLA-450M, OpenVLA-7B, and OpenVLA-OFT-7.5B on the ten tasks of

	No Goal Achieved ($\mathcal{A}(S_T) = \emptyset$)	Some Goal Achieved ($\mathcal{A}(S_T) \neq \emptyset$)
I_{orig} **Succeeded**	**Grounded.** Removing language eliminated task success.	**Memorization.** Policy succeeded without language; driven by scene priors.
I_{orig} **Failed**	**Uninformative.** Policy incompetent on this scene; excluded from grounding rate.	**Spurious Success.** Policy achieved a goal without language where original prompt failed.

Table 2: The destructive-condition (`blank`, `nonsense`) 2×2 evaluation matrix.

	Follow ($k_{\text{wrong}} \in \mathcal{A}(S_T)$)	Ignore ($k^* \in \mathcal{A}(S_T)$)	Fail (Neither Goal)
I_{orig} **Succeeded**	*Instruction-following.* Switched to newly instructed goal when asked.	*Scene-prior override.* Defaulted to original task despite new prompt.	*Confused.* Misleading prompt broke policy without redirection.
I_{orig} **Failed**	*Instruction-following.* Completed new goal despite failing baseline prompt.	*Noisy execution.* Achieved original goal unreliably.	*Uninformative.* Failed task under both conditions.

Table 3: The misleading-condition (`wrong_object`, `wrong_action`, `wrong_task`) 3×2 evaluation matrix.

`LIBERO-Goal` in MuJoCo. All tasks share an identical initial scene layout and object inventory, ensuring that the instruction is the sole distinguishing input.

5.2 Scene-Fixed Protocol

For each (task, episode) pair, we hold the initial simulator state S_0 strictly constant across all instruction conditions: $S_0(I_{\text{orig}}) \equiv S_0(I_{\text{pert}})$. We audit this invariant by hashing simulator state arrays (`qpos`, `qvel`) immediately after environment reset. As detailed in Section B, un-audited evaluations frequently violate this invariant due to simulator generator drift across process resets.

5.3 Data Collection and Implementation

Each rollout logs a structured record containing model checkpoint hash, task ID, condition, seed, initial-state hash, achieved task set $\mathcal{A}(S_T)$, and end-effector kinematics. Rollouts run for $T = 300$ steps under seeds 7 and 42. Simulator evaluations execute sequentially with one EGL rendering context per process to prevent rendering context crashes.

6 Results and Analysis

6.1 RQ1: Language is Causally Load-Bearing and Scale-Invariant

Table 4 presents Task Success Rate (TSR), Causal Sensitivity Score (CSS), and Original-Action Rate (OAR) under structural and semantic perturbations.

Destructive conditions (`blank`, `nonsense`) collapse task success across all architectures. SmolVLA-450M drops to 0.0% success (CSS $=$ 1.00). OpenVLA-7B drops to 0.0% under `blank` (CSS $=$ 1.00). OpenVLA-OFT retains a minor residue (12.5% under `blank`, CSS $=$ 0.87), which we attribute to visual shortcuts enabled by its wrist camera and proprioception inputs. Crucially, causal sensitivity is flat across model scale: CSS remains near 1.00 from 0.45B to 7.5B parameters.

6.2 RQ2: Information Structure Confounds Word-Class Grounding

Prior studies observe that object swaps collapse performance while verb swaps are tolerated, concluding that VLAs bind nouns strictly but ignore verbs. We find that this conclusion is confounded by the benchmark's information structure.

We compute the conditional task entropy $H(\text{task} \mid W)$ over `LIBERO-Goal`'s ten task instructions under a uniform task prior ($H(\text{task}) = \log_2 10 = 3.32$ bits):

$$\begin{aligned} H(\text{task} \mid \text{noun set}) &= 0.00 \text{ bits}, \\ H(\text{task} \mid \text{verb set}) &= 1.55 \text{ bits}. \end{aligned} \tag{3}$$

All ten noun sets are unique across tasks, resolving all 3.32 bits of task identity. In contrast, the verb `put` appears in six of the ten tasks, leaving 1.55 bits unresolved (Figure 1). Conditional on knowing the nouns, the verb contributes *exactly zero bits* of task information.

This information redundancy has two major implications: 1. Ignoring the verb is informationally optimal on `LIBERO-Goal`. 2. Standard verb substitution probes are uninformative by construction: 7 of the 10 verb swaps replace `put` with `push`, leaving the noun set and task identity unchanged.

Model	Baseline	Blank	Nonsense	Wrong Action	Wrong Object	Wrong Task	Repeated
SmolVLA (0.45B)	78.5%±2.5	0.0% (1.00)	0.0% (1.00)	64.5%±0.5	9.3%±2.1	0.0%	10.0%±2.0
OpenVLA (7B)	71.0%±1.0	0.0% (1.00)	0.0% (1.00)	59.0%	10.7%±0.7	0.0%	58.5%±0.5
OpenVLA-OFT (7.5B)	97.5%±1.5	12.5%±2.5 (0.87)	14.0%±1.0 (0.86)	94.5%±0.5	11.4%	0.0%	96.5%±1.5
episodes per cell	200	200	200	200	140	200	200

Table 4: TSR under structural and semantic perturbations on `LIBERO-Goal`, with CSS in parentheses for destructive conditions. Episode counts differ by condition and model, so there is no single N: `original` $n = 200$, `blank` $n = 200$, `nonsense` $n = 200$, `wrong_action` $n = 200$, `wrong_object` $n = 140$, `wrong_task` $n = 200$, `repeated` $n = 200$. Seeds 42, 7. All models pass the scene-fixed check on every comparable episode.

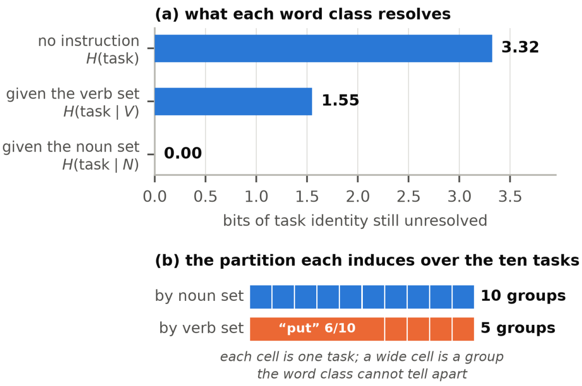

Figure 1: `LIBERO-Goal` information structure. **(a)** Nouns resolve all 3.32 bits of task identity; verbs leave 1.55 bits unresolved. **(b)** Verbs partition six tasks into a single uninformative `put` cluster.

Condition	bits	TSR	Δ [95% CI]	p
OpenVLA (7B)				
`original`	0.00	71.0%	0.0	–
`verb_dropped`	0.00	75.0%	+4.0 [−3.5, +11.0]	0.36
`nouns_masked`	3.32	6.0%	−65.0 [−71.5, −58.0]	$< 10^{-10}$
`blank`	3.32	0.0%	−71.0 [−77.0, −64.5]	$< 10^{-10}$
OpenVLA-OFT (7.5B)				
`original`	0.00	97.5%	0.0	–
`verb_dropped`	0.00	95.5%	−2.0 [−5.0, +1.0]	0.34
`nouns_masked`	3.32	24.0%	−73.5 [−79.5, −67.5]	$< 10^{-10}$
`blank`	3.32	12.5%	−85.0 [−89.5, −80.0]	$< 10^{-10}$

Table 5: Word-class removal test ($n = 200$ per cell). Coverage: OpenVLA (7B) on 10 of 10 tasks; OpenVLA-OFT (7.5B) on 10 of 10 tasks. The ablation conditions are generated in the same process as `original`, so the Δ column is a within-scene contrast.

6.3 The Word-Class Removal Test

To test this entropy account, we perform a removal test: rather than substituting words, we drop word classes outright. Deleting the verb (0 bits destroyed) should leave performance unchanged, whereas masking nouns (3.32 bits destroyed) should collapse performance.

The empirical results confirm our prediction (Table 5, Figure 2). Deleting the verb changes success by only 2.0 points on OpenVLA-OFT (McNemar $p = 0.34$, 10 changed outcomes out of 200 matched pairs, consistent with random chance). Masking nouns costs 73.5 points and flips 148 pairs to failure. Two edits of similar edit distance differ by two orders of magnitude in impact because of their information content.

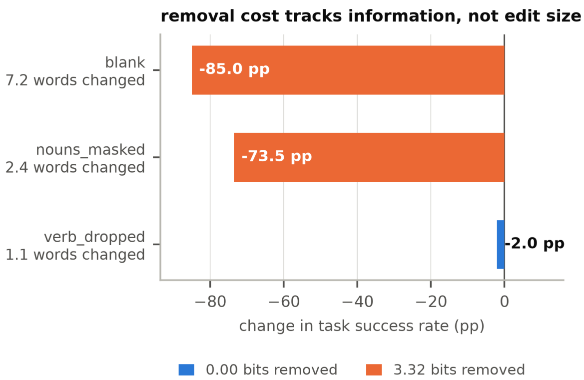

Figure 2: Removal test on OpenVLA-OFT. Deleting the verb costs 2.0 points ($p = 0.34$), while masking nouns costs 73.5 points.

6.4 Re-evaluating Word-Class Asymmetry: Antonym vs. Near-Synonym Split

We separate the ten verb substitutions into the 3 genuine antonym pairs (`open`↔`close` twice, `turn on`↔`turn off` once) and the 7 near-synonym control pairs (Table 6, Figure 3).

Splitting the probe reverses the conclusion for OpenVLA-7B. Evaluating the suite-wide `wrong_action` probe yields a mild 12.0-point drop. However, splitting the probe reveals that OpenVLA-7B drops 31.7 points on true antonyms ($p < 10^{-5}$), while dropping only 3.6 points on near-synonyms ($p = 0.52$). OpenVLA-7B actively reads and grounds action verbs when the prompt contains true task contrast; suite-wide probes mask this by averaging real effects with uninformative controls.

Model	antonym		near-synonym	
	ΔTSR	p	ΔTSR	p
SmolVLA (0.45B)	−21.7 pp	10^{-3}	−10.7 pp	0.02
OpenVLA (7B)	−31.7 pp	10^{-6}	−3.6 pp	0.52
OpenVLA-OFT (7.5B)	−5.0 pp	0.45	−2.1 pp	0.51

Table 6: TSR drop under verb substitution split by task identity change (p-value from McNemar test over matched pairs).

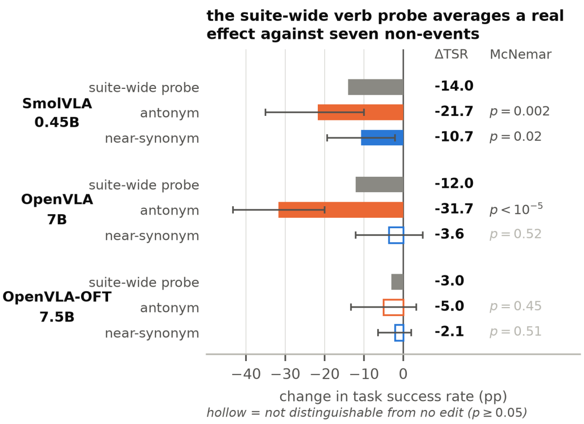

Figure 3: Verb substitution split into antonyms ($n = 60$) versus near-synonyms ($n = 140$). OpenVLA-7B drops 31.7 points on antonyms ($p < 10^{-5}$).

6.5 Paraphrase Robustness and Kinematics

Table 7 reports PRIDE paraphrase scores. Robustness scales with parameter count: SmolVLA scores PRIDE 2.7, OpenVLA-7B scores 33.3, and OpenVLA-OFT achieves 65.8. All models follow the same vulnerability ordering: object paraphrases degrade performance least, while compositional paraphrases degrade performance most.

Kinematic analysis (t_{div}, threshold 5 cm) confirms these findings: under `blank` instructions, trajectories depart from canonical paths within 1–2 steps. Under `wrong_action`, trajectories follow canonical paths for 6–11 steps before diverging, matching the timing expected when nouns specify target approach location and verbs specify manipulation manner.

Model	Paraphrase Axis	n	scenes	TSR	ΔTSR (pp)	PRIDE
SmolVLA (0.45B)						
SmolVLA (0.45B)	Object	259	20	29.73%	−48.77	28.7
SmolVLA (0.45B)	Action	870	20	7.24%	−71.26	4.8
SmolVLA (0.45B)	Compositional	2963	20	1.79%	−76.71	1.5
SmolVLA (0.45B)	**All axes**	4092	24	4.72%	−73.78	2.7
OpenVLA (7B)						
OpenVLA (7B)	Object	150	150	48.00%	−23.00	36.3
OpenVLA (7B)	Action	150	150	60.67%	−10.33	51.8
OpenVLA (7B)	Compositional	150	150	24.00%	−47.00	20.2
OpenVLA (7B)	**All axes**	450	150	44.22%	−26.78	33.3
OpenVLA-OFT (7.5B)						
OpenVLA-OFT (7.5B)	Object	150	150	78.00%	−19.50	72.5
OpenVLA-OFT (7.5B)	Action	150	150	90.67%	−6.83	86.3
OpenVLA-OFT (7.5B)	Compositional	150	150	54.00%	−43.50	50.3
OpenVLA-OFT (7.5B)	**All axes**	450	150	74.22%	−23.28	65.8

Table 7: Paraphrase robustness scored via PRIDE.

7 Discussion and Conclusion

We evaluated three VLA architectures under a scene-fixed causal protocol to determine the nature of language grounding failures. Our findings establish three key conclusions:

1. **Language is causally necessary:** Completely removing language collapses task success across all scales, disproving the notion that VLAs operate as purely visual policies on shared-scene benchmarks. 2. **Benchmark information redundancy creates apparent verb blindness:** In `LIBERO-Goal`, nouns uniquely determine task identity. Models ignore verbs because verbs are redundant, not because models are incapable of verb grounding. 3. **VLAs ground verbs when prompts are contrastive:** On true verb antonyms, OpenVLA-7B exhibits strong verb sensitivity (31.7-point drop, $p < 10^{-5}$).

We recommend that future VLA benchmarks measure and report instruction set information content before drawing word-class grounding conclusions.

Limitations

Our study has several scope bounds: (1) **Single benchmark suite:** Causal isolation relies on `LIBERO-Goal`'s shared initial scene. (2) **Antonym task coverage:** Only three tasks in `LIBERO-Goal` afford true verb antonyms ($n = 60$ episodes across 3 task templates). (3) **Simulation scope:** All experiments take place in MuJoCo simulation without real-robot deployment. (4) **Uniform entropy prior:** Information calculations assume a uniform task distribution, which may differ from model pre-training data distributions.

Ethics and Reproducibility

This work uses simulated environments and open-source model checkpoints. We release per-run manifests (seed, environment locks, state hashes) to ensure full computational reproducibility.

References

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, et al. 2023. Rt-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*.

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, et al. 2022. Rt-1: Robotics transformer for real-world control at scale. *arXiv preprint arXiv:2212.06817*.

Open X-Embodiment Collaboration, Abhishek Padalkar, Acorn Pooley, et al. 2023. Open x-embodiment: Robotic learning datasets and rt-x models. *arXiv preprint arXiv:2310.08864*.

Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, et al. 2023. Palm-e: An embodied multimodal language model. *arXiv preprint arXiv:2303.03378*.

Senyu Fei, Siyin Wang, Junhao Shi, Zihao Dai, Jikun Cai, Pengfang Qian, Li Ji, Xinzhe He, Shiduo Zhang, Zhaoye Fei, Jinlan Fu, Jingjing Gong, and Xipeng Qiu. 2025. Libero-plus: In-depth robustness analysis of vision-language-action models. *arXiv preprint arXiv:2510.13626*.

Yash Goyal Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2017. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6904–6913.

Bryce Grant, Xijia Zhao, and Peng Wang. 2026. Not all features are created equal: A mechanistic study of vision-language-action models. *arXiv preprint arXiv:2603.19233*.

Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R Bowman, and Noah A Smith. 2018. Annotation artifacts in natural language inference data. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 107–112.

Yuchen Hou and Lin Zhao. 2026. Langgap: Diagnosing and closing the language gap in vision-language-action models. *arXiv preprint arXiv:2603.00592*.

Chanyoung Kim, Minwoo Kim, Minseok Kang, Hyunwoo Kim, and Dahuin Jung. 2026. Libero-para: A diagnostic benchmark and metrics for paraphrase robustness in vla models. *arXiv preprint arXiv:2603.28301*.

Moo Jin Kim, Chelsea Finn, and Percy Liang. 2025. Fine-tuning vision-language-action models: Optimizing speed and success. *arXiv preprint arXiv:2502.19645*.

Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. 2024. Openvla: An open-source vision-language-action model. In *Proceedings of the 8th Conference on Robot Learning (CoRL)*. ArXiv:2406.09246.

Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. 2023. Libero: Benchmarking knowledge transfer for lifelong robot learning. *arXiv preprint arXiv:2306.03310*.

Tom McCoy, Ellie Pavlick, and Tal Linzen. 2019. Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 3428–3448.

Yiran Pang, Yiheng Zhao, Zhuopu Zhou, Tingkai Hu, and Ranxin Hou. 2025. Is openvla truly robust? a systematic evaluation of positional robustness. In *Proceedings of the 14th International Joint Conference on Natural Language Processing (IJCNLP), Short Papers*.

Mustafa Shukor, Dana Aubakirova, Francesco Capuano, Peter Kooijmans, Stefano Palma, Adil Zouitine, Michel Aractingi, Caroline Pascal, Martino Russi, Andres Marafioti, Simon Alibert, Matthieu Cord, Thomas Wolf, and Remi Cadene. 2025. Smolvla: A vision-language-action model for affordable and efficient robotics. *arXiv preprint arXiv:2506.01844*.

Zhihao Zhan, Yuhao Chen, Jiaying Zhou, Qinhan Lyu, Hao Liu, Keze Wang, Liang Lin, and Guangrun Wang. 2026. Stable language guidance for vision-language-action models. *arXiv preprint arXiv:2601.04052*.

Haochen Zhang and Yonatan Bisk. 2026. Flatness preserves instruction following in vision-language-action models. *arXiv preprint arXiv:2606.23641*.

Xueyang Zhou, Yangming Xu, Guiyao Tie, Yongchao Chen, Guowen Zhang, Duanfeng Chu, Pan Zhou, and Lichao Sun. 2025. Libero-pro: Towards robust and fair evaluation of vision-language-action models beyond memorization. *arXiv preprint arXiv:2510.03827*.

A Full Per-Cell Results

Table 8 provides per-cell episode counts and 95% Wilson confidence intervals for all RQ1 conditions. Table 9 reports kinematic divergence metrics.

B Scene-Fixed Invariant Audit

Standard LIBERO reset logic advances an internal generator across calls. As a result, sequential conditions run in separate processes silently receive different initial object poses. We enforce per-task process execution and validate state hashes (`qpos`, `qvel`) to guarantee strict scene matching across conditions.

Model	Condition	n	TSR (95% CI)	CSS
SmolVLA (0.45B)	`original`	200	78.5% [72.3, 83.6]	–
SmolVLA (0.45B)	`blank`	200	0.0% [0.0, 1.9]	1.00
SmolVLA (0.45B)	`nonsense`	200	0.0% [0.0, 1.9]	1.00
SmolVLA (0.45B)	`wrong_action`	200	64.5% [57.7, 70.8]	–
SmolVLA (0.45B)	`wrong_object`	140	9.3% [5.5, 15.2]	–
SmolVLA (0.45B)	`wrong_task`	200	0.0% [0.0, 1.9]	–
SmolVLA (0.45B)	`repeated`	200	10.0% [6.6, 14.9]	–
OpenVLA (7B)	`original`	200	71.0% [64.4, 76.8]	–
OpenVLA (7B)	`blank`	200	0.0% [0.0, 1.9]	1.00
OpenVLA (7B)	`nonsense`	200	0.0% [0.0, 1.9]	1.00
OpenVLA (7B)	`wrong_action`	200	59.0% [52.1, 65.6]	–
OpenVLA (7B)	`wrong_object`	140	10.7% [6.6, 16.9]	–
OpenVLA (7B)	`wrong_task`	200	0.0% [0.0, 1.9]	–
OpenVLA (7B)	`repeated`	200	58.5% [51.6, 65.1]	–
OpenVLA-OFT (7.5B)	`original`	200	97.5% [94.3, 98.9]	–
OpenVLA-OFT (7.5B)	`blank`	200	12.5% [8.6, 17.8]	0.87
OpenVLA-OFT (7.5B)	`nonsense`	200	14.0% [9.9, 19.5]	0.86
OpenVLA-OFT (7.5B)	`wrong_action`	200	94.5% [90.4, 96.9]	–
OpenVLA-OFT (7.5B)	`wrong_object`	140	11.4% [7.2, 17.8]	–
OpenVLA-OFT (7.5B)	`wrong_task`	200	0.0% [0.0, 1.9]	–
OpenVLA-OFT (7.5B)	`repeated`	200	96.5% [93.0, 98.3]	–

Table 8: Full RQ1 per-cell success rates with 95% Wilson confidence intervals.

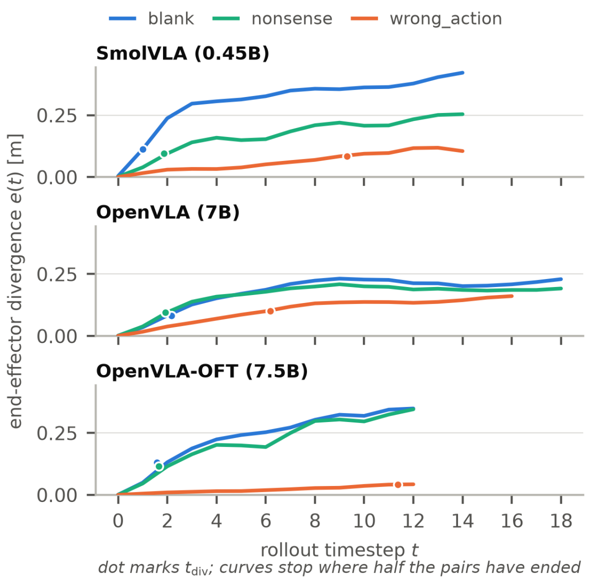

Figure 4: End-effector displacement over rollout timesteps.

C PRIDE Metric Formulation

PRIDE (Kim et al., 2026) weights paraphrase success by distance from canonical instructions: $\mathrm{PD}_i = 1 - 0.5(\mathrm{SK}_i + \mathrm{ST}_i)$ and $\mathrm{PRIDE} = 100 \sum_i s_i \mathrm{PD}_i / \sum_i \mathrm{PD}_i$.

D Kinematic Divergence Analysis

Figure 4 illustrates end-effector divergence $e(t)$ over time step t.

E Initial Pilot Experiments on SmolVLA

In preliminary experiments on SmolVLA-450M using a 4-task subset of `LIBERO-Object`, blank instructions dropped success from 100% (8/8) to 0% (0/8), while object swaps dropped success to 0% (0/8), confirming early directional sensitivity before expanding to the full 10-task `LIBERO-Goal` suite.

Model	Condition	pairs	t_{div} (samples)	e_{10} [m]
SmolVLA (0.45B)	blank	84	1.00	0.3548
SmolVLA (0.45B)	nonsense	84	1.87	0.2197
SmolVLA (0.45B)	para_action	10	2.00	0.1972
SmolVLA (0.45B)	para_compositional	13	1.85	0.2086
SmolVLA (0.45B)	para_object	13	4.69	0.1383
SmolVLA (0.45B)	repeated	84	2.40	0.1737
SmolVLA (0.45B)	wrong_action	77	9.31	0.0838
SmolVLA (0.45B)	wrong_task	84	1.64	0.3278
SmolVLA (0.45B)	wrong_object	56	1.96	0.1785
OpenVLA (7B)	blank	73	2.16	0.2315
OpenVLA (7B)	nonsense	73	1.92	0.2087
OpenVLA (7B)	nouns_masked	73	3.11	0.2015
OpenVLA (7B)	repeated	65	6.35	0.1082
OpenVLA (7B)	verb_dropped	68	7.87	0.0764
OpenVLA (7B)	wrong_action	68	6.18	0.1355
OpenVLA (7B)	wrong_task	73	2.25	0.2992
OpenVLA (7B)	wrong_object	48	2.75	0.1920
OpenVLA-OFT (7.5B)	blank	55	1.58	0.3236
OpenVLA-OFT (7.5B)	nonsense	54	1.65	0.3043
OpenVLA-OFT (7.5B)	nouns_masked	55	2.67	0.2751
OpenVLA-OFT (7.5B)	para_action	50	11.70	0.0339
OpenVLA-OFT (7.5B)	para_compositional	55	6.58	0.1474
OpenVLA-OFT (7.5B)	para_object	51	10.90	0.0582
OpenVLA-OFT (7.5B)	repeated	43	12.02	0.0234
OpenVLA-OFT (7.5B)	verb_dropped	53	12.13	0.0206
OpenVLA-OFT (7.5B)	wrong_action	51	11.37	0.0289
OpenVLA-OFT (7.5B)	wrong_task	55	1.64	0.3127
OpenVLA-OFT (7.5B)	wrong_object	35	1.89	0.3173

Table 9: Kinematic divergence step t_{div} and end-effector displacement e_{10}.